CSE 578: Course Project Final Report

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I. Goals and Business Objective

THE primary goal of the UVW marketing team is to understand the relationship between an individual's factors and their income level, as UVW has previously determined that an individual of income level $<= \$50k$ is their key demographic in marketing towards. This is because UVW is undergoing an effort to bolster their enrollment, and they believe that increasing marketing towards these individuals could be successful. To do this, UVW wants to build an application that can "predict the income of an individual based on different values of the input parameters so that they can tailor their marketing efforts when reaching out to the individuals."

XYZ Corporation is responsible for providing effective marketing profiles and determining the most important variables for the UVW marketing team to consider in how they decide which individuals to market to.

II. Assumptions

- The dataset has a significant class imbalance, which might reflect the true population distribution. However, without knowledge of future UVW data collection or the true population, arguing if results stem from sampling bias or not should be ignored for now.
- Relationships in the database should be considered as correlations rather than real-world causations. Determining causation should be done in future experiments or projects by the UVW marketing team.

III. User Stories

A. As a member of the UVW marketing team, I want to know the average importance of each measurable factor of an individual in determining their income level.

This user story serves as a general overview of determining which factors should be considered moving forward for UVW. For example, if the "Hours Worked" feature ends up not having much of an importance or weight in determining an individual's income level, then it could be better left out of the picture. On the other hand, if it does turn out to be extremely important, then it should definitely be considered by the UVW team.

B. As a member of the UVW marketing team, I want to know if the individuals who meet/don't meet the target income level are similar or dissimilar between each other.

This user story helps to determine the homogenity of the target groups. For example, if it ends up being that the distribution of people who meet the target income level is such that everyone is very different from each other, then that means the marketing team would have to consider people among a very wide range of features.

C. As a member of the UVW marketing team, I want to know if there are any relationships between values of categorical features of the data set.

This user story gives a more in-depth comparison for a given feature. For example, when considering marital status of an individual, finding if it is more important for someone to be single or married in determining if they meet the $<= \$50k$ income target could help in deciding how to specifically market to people given their factors.

D. As a member of the UVW marketing team, I want to view the relationships of the top factors in determining an individual's income level. For example, gender vs. income.

This user story provides the most actionable visualizations by targeting the most important features and comparing them in their actual relationship compared to income level.

E. As a member of the UVW marketing team, I want to construct the most typical individual who meets the target income level.

This user story would help build the UVW marketing team a representative marketing profile to determine their biggest target audience/demographic, if there is one significant enough to exist.

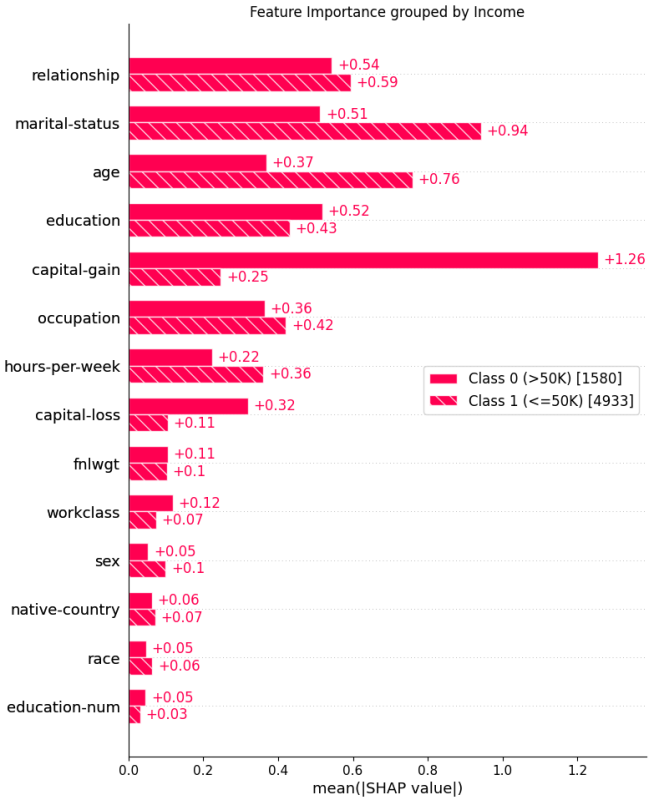

Fig. 1. Table of feature importance grouped by income.

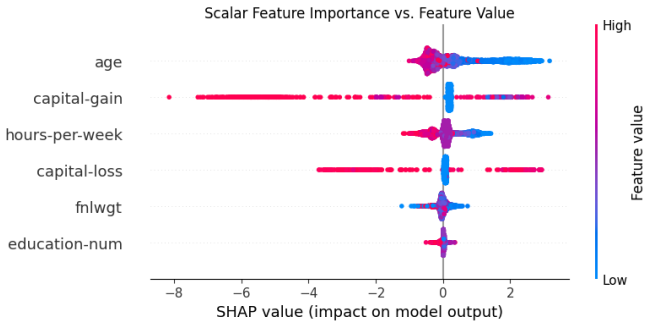

Fig. 2. Table of scalar feature importance vs. feature value (low to high, numerically)

IV. Visualizations

A. *As a member of the UVW marketing team, I want to know the average importance of each measurable factor of an individual in determining their income level.*

In order to calculate the importance of each factor in determining income level, SHAP values are calculated based a the XGBoost model trained on the data. SHAP values are used to quantify how much a feature contributes to a model's prediction [1]. It is commonly used in classifier-explanation applications for its "theoretically robust way to interpret predictions," although it does have some limitations of introducing some biases and does not necessarily explain causal relationships [2]. Given UVW's goal of creating a predictor for income level, however, understanding feature importance will likely be very helpful. In Figure 1, each feature is grouped by target income on its importance in calculating the class. Higher values mean that the feature is more important in determining the class. For example, we can see that age is a high-impact feature in determining $<= \$50k$ income, but less impactful in determining $> \$50k$ income. We also generally see that the most impactful features for predicting of $<= \$50k$ income are marital-status, age, relationship, education, etc. Figure 2 demonstrates how the range of scalar values impacts feature importance. For example, we can conclude that low values of age/young people have high correlation to positive SHAP value (good predictor of $<= \$50k$ income), and high values of age/old people have decent correlation to negative SHAP value (good predictor of $> \$50k$ income). Notably, high capital gain is generally a good indicator of $> \$50k$ income, and this can be observed in both Figure 1 and Figure 2.

B. *As a member of the UVW marketing team, I want to know if the individuals who meet/don't meet the target income level are similar or dissimilar between each other.*

The primary purpose of this visualization is to see if any kind of structure in similarity exists between individuals of $<= \$50k$ (yellow) income and those of $> \$50k$ (purple) income. Since each instance has a high dimension considering all the different features, t-SNE is used to decompose them into a 2D visualization, producing a 2D scatterplot, Figure 3. We can observe some clusters among yellow points, but purple points seem to be mostly randomly dispersed around the graph. This could indicate that there indeed exists clusters of $<= \$50k$ individuals that have high intra-cluster similarity, but not necessarily that all individuals are very alike, as there are notable differences between the yellow clusters. Only one notable cluster seems to exist for $> \$50k$ individuals, and interestingly, it is mixed in very well with a seemingly equal number of yellow dots. This could indicate that certain groups of individuals of differing income classes can still be very similar across their features. However, since our dataset has significant class imbalence, a more uniform data set could show different results.

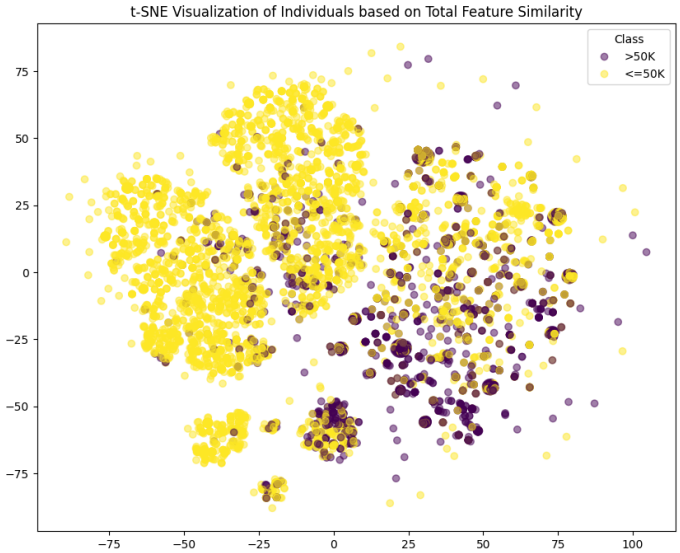

Fig. 3. t-SNE Visualization of Individuals based on total feature similarity. Yellow is individuals of income $<= \$50k$. Purple is individuals of income $> \$50k$.

C. As a member of the UVW marketing team, I want to know if there are any relationships between values of categorical features of the data set.

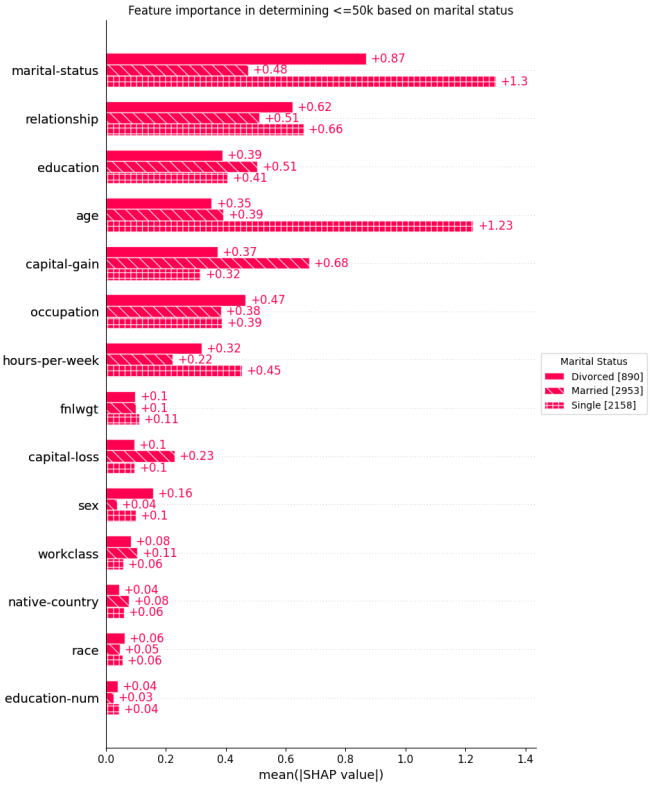

Fig. 4. Feature importance in determining $<= \$50k$ based on marital status

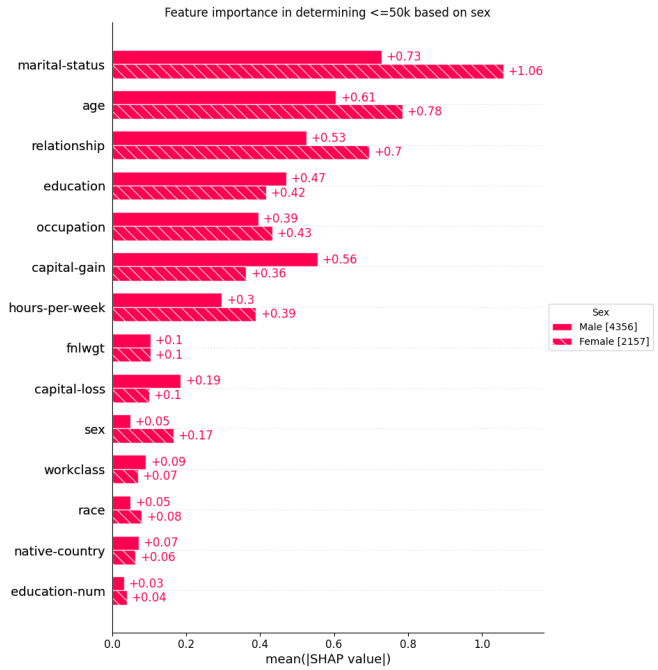

Fig. 5. Feature importance in determining $<= \$50k$ based on sex

We return to SHAP values, but this time, we consider grouping by possible values of certain features. For Figure 4 and Figure 5, we consider categorical features. In Figure 4, we see that for single individuals, there is an extremely high importance of age in determining of they have income $<= \$50k$, while it is not as important for divorced and married individuals. Interestingly, capital gain is very important for married individuals, relative to the other marital status types. In Figure 5, we can see some significant differences in feature importance between male and female individuals concerning marital status, age, relationship, hours-per-week worked, and capital gain.

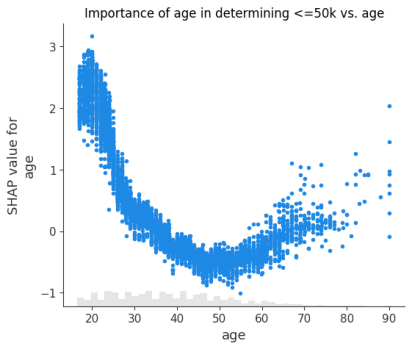

Fig. 6. Importance in determining $<= \$50k$ vs. age

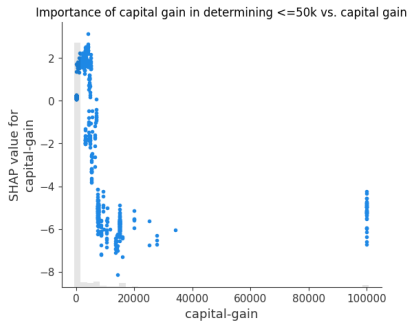

Fig. 7. Importance in determining $<= \$50k$ vs. capital gain

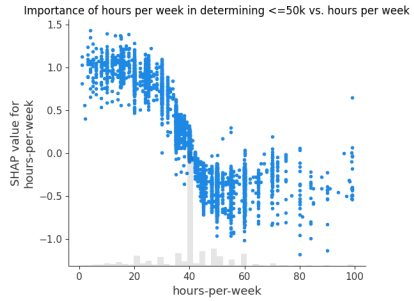

Fig. 8. Importance in determining $<= \$50k$ vs. hours per week

To consider scalar values, we use a scatter plot indicating SHAP value for each instance in the dataset for Figures 6-8. Interestingly, we observe very similar curves and distribution among the three graphs of age, capital gain, and hours-per-week worked. Low values tend to indicate high value in predicting an individual of $< \$50k$ income. SHAP value then drops sharply as it reaches what is likely the median value in the range for the feature. For example, considering age, SHAP value steeply falls from the range of 20 years to 40 years, then slightly goes up but has a much looser positive correlation. This could indicate that as the individual gets older, age becomes less important and less accurate to use as a feature for determining their income level.

D. As a member of the UVW marketing team, I want to view the relationships of the top factors in determining an individual's income level. For example, gender vs. income.

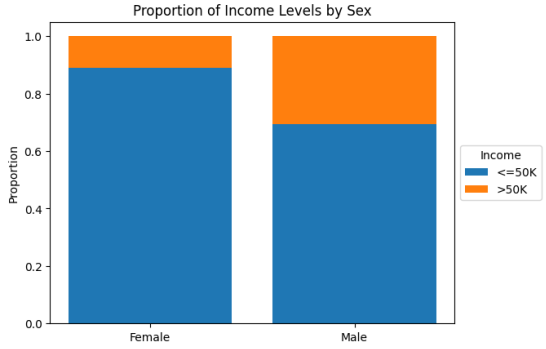

Fig. 9. Proportion of income levels by sex

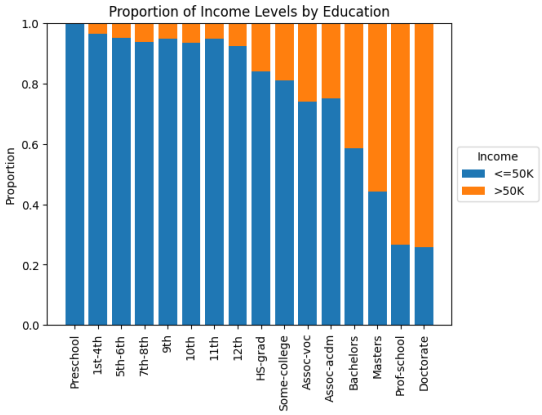

Fig. 10. Proportion of income levels by education

We move away from SHAP values to concentrate on the actual instances of the data set. This allows us to find insights on the actual distribution of the sample population, rather than insights on predictability performance. Figures 9 through 11 are stacked bar charts that represent the proportion of individuals of the target $<= \$50k$ income level and those of $> \$50k$ income level grouped by the different categories of a categorical feature. Figure 9 shows that in the sample, females are more likely to have lower income than males. Figure 10 shows that the proportion of higher income level by education has very small increments from 1st to 12th grade, and has a noticible increase at the HS-grad level. The proportion of higher income levels starts to sharply increase as it progresses from associate's to bachelor's, master's, and eventual doctoral degrees. Figure 11 follows a similar format but is grouped by occupation. "Priv-House-Serv" represents "Private household services," which group occupations like household cleaning, maids, etc., and has the highest proportion of individuals of $<= \$50k$ income level. The other trade-related occupations generally seem to have higher proportions of individuals of $<= \$50k$ income level than occupations like management, professional specialties, etc. Figure 12 is a histogram of proportion of income level based on age buckets, and generally, the proportion of $<= \$50k$ individuals are highest at young ages, lowest in the median at around 40-50 years old, and then starts to rise again past 60.

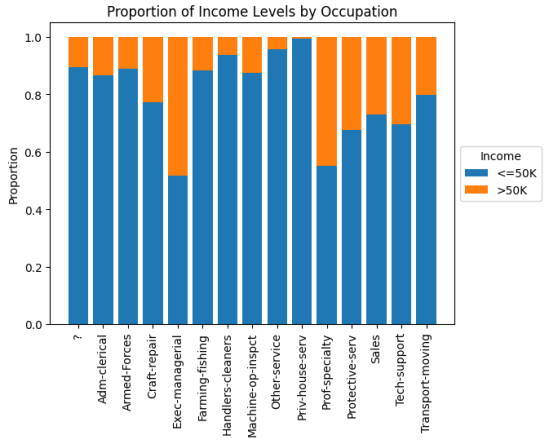

Fig. 11. Proportion of income levels by occupation

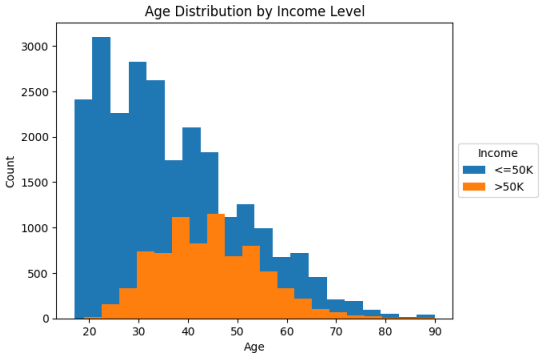

Fig. 12. Proportion of income levels by income level

Marketing profile for <=50K Individuals from mode and median values

Feature	Value
age	34
workclass	Private
education	HS-grad
marital-status	Never-married
occupation	Adm-clerical
relationship	Not-in-family
race	White
sex	Male
capital-gain	0
capital-loss	0
hours-per-week	40
native-country	United-States

Fig. 13. Marketing profile for $<= \$50k$ individuals from mode and mean values

Highest count actual profile for <=50K Individuals, Count: 16

Feature	Value
age	33
workclass	Private
education	HS-grad
marital-status	Married-civ-spouse
occupation	Craft-repair
relationship	Husband
race	White
sex	Male
capital-gain	0
capital-loss	0
hours-per-week	40
native-country	United-States

Fig. 14. Highest count actual profile for $<= \$50k$ individuals, Count: 16

E. As a member of the UVW marketing team, I want to construct the most typical individual who meets the target income level.

Constructing the marketing profile was done by taking the mode value of categorical features and median value of scalar features of individuals who are of income level $<= \$50k$. This would be helpful for UVW in marketing as it gives a general overview of the kind of individual that they might want to target their marketing towards. Figure 13 shows that the most common individual meeting the target income level is most likely to be around 34 years old, graduated high school as their highest education level, is male, single, and works 40 hours per week.

Figure 14 represents the highest count profile found in the dataset for individuals who are under $<= \$50k$. It is similar in purpose to the marketing profile, but is an aggregation and represents actual instances of the data set.

F. Summary of Findings

Overall, these visualizations show that there is a clear structure for predicting the income level of an individual based on the features used, and that different features have varying importance level based on individual factors/the individual's actual income level. The most five important features for predicting $<= \$50k$ individuals are marital status, age, relationship,

education, and occupation. Some factors may be highly correlation with each other, such as age importance for an individual who is single (see Figure 4). It is important to keep in mind that the proportion of individuals grouped by income may be skewed due to the class imbalance of income level in the data set.

V. Questions

A. Should we use a predictor model in producing this report? And if so, which one, and why?

UVW's goal is to produce an applicatioon that can predict income level based on an individual's factors, so even though the task of building a successful predictor is not the sole purpose of this report, doing so will help explain which factors will be useful for UVW to consider in future marketing plans. For example, if they want to do data collection, but they have limited factors to choose from.

For the choice of predictor model, my first choice was to use XGBoost. This is because XGBoost is generally better performing than simpler ensemble models like Random Forest, and includes "an array of techniques to handle missingness, imbalanced data via weighting and large datasets in addition to other powerful software-level optimizations" [3]. Additionally, when comparing XGBoost, Random Forest, and Logistic Regression, XGBoost indeed performed the best among all metrics.

Model	PR-AUC	F1 Score
XGBoost	0.97	0.91
Random Forest	0.94	0.88
Logistic Regression	0.95	0.89

TABLE I
PR-AUC and F1 scores for different models [4]

XGBoost also had the advantage of being able to handle categorical data by-default, so it was much easier to implement. The XGBoost library also comes with its own visualizers and functions for things like feature importance, but I elected to use the SHAP library for fulfilling that purpose.

B. For data preprocessing, how should the database be processed in order to feed it into models like t-SNE that do not, by-standard, accept categorical data?

This is one of the issues that I encountered both when implementing t-SNE and for testing logisistic regression and random forest models, though I only used the XGBoost model for the calculation of SHAP values. The solution was to use $pd.get_dummies()$ on the categorical columns to do a one-hot encoding for them. However, this could be responsible for the long processing time of the t-SNE graph. Additionally, it's indeterminant if doing this process led to reduced accuracy in the resultant graph.

C. How should visualizations be done when considering categorical features vs. scalar features?

My intuition was to primarily use bar charts for categorical data. For considering scalar features, the SHAP library provided a beeswarm plot, which is is a multivariate scatter plot similar to a bar chart in that it is grouped by feature, but each point represents the instance. Its distance along the x-axis represents the SHAP value/importance, and its color represents the actual feature value. For later figures, I used a traditional scatter plot where each plot only needed to consider one feature. When going to the proportions, I decided to use a histogram for age.

VI. Not doing/TODO

The first action that should be taken would be to generate these visualizations on a data set that has an even class balance. Indeed, there may be a class imbalance in the real world and the data set for this report may accurately represent the real-world population, but I think it still is important to compare the resulting visualizations and results to identify if XGBoost, or UVW's chosen prediction model, generates biases on specific features due to the class imbalance. I would recommend XGBoost, still.

Additional visualizations could be created to identify important relationships and could stem from the examples of Figures 4-8. These visualizations reveal some insights that may not be obvious, such as how the age feature becomes vastly more important for single individuals than married/divorced individuals when considering the prediction of $<= \$50k$ income.

References

[1] S. Lundberg, "An introduction to explainable AI with Shapley values — SHAP latest documentation," Readthedocs.io, 2018. https://shap.readthedocs.io/en/latest/example_notebooks/overviews/An%20introduction%20to%20explainable%20AI%20with%20Shapley%20values.html

[2] Y. Takefuji, "Beyond XGBoost and SHAP: Unveiling true feature importance," Journal of Hazardous Materials, vol. 488, p. 137382, May 2025, doi: https://doi.org/10.1016/j.jhazmat.2025.137382.

[3] M. Wiens, A. Verone-Boyle, N. Henscheid, J. T. Podichetty, and J. Burton, "A Tutorial and Use Case Example of the eXtreme Gradient Boosting (XGBoost) Artificial Intelligence Algorithm for Drug Development Applications," Clinical and Translational Science, vol. 18, no. 3, Mar. 2025, doi: https://doi.org/10.1111/cts.70172.

[4] tomasgdejesus, "CSE578-Project/mlmodels.ipynb at main · tomasgdejesus/CSE578-Project," GitHub, 2026. https://github.com/tomasgdejesus/CSE578-Project/blob/main/mlmodels.ipynb (accessed Apr. 22, 2026).